Assignment No.04 (Travelling Salesperson Problem)

- Code:

```
import random
import numpy as np

def distance_matrix(n):
    return np.random.randint(10, 100, size=(n, n))

def total_distance(route, distances):
    return sum(distances[route[i], route[i + 1]] for i in
range(len(route) - 1)) + distances[route[-1], route[0]]

def generate_neighbors(route):
    neighbors = []
    for i in range(len(route)):
        for j in range(i + 1, len(route)):
            new_route = route[:]
            new_route[i], new_route[j] = new_route[j],
new_route[i]
            neighbors.append(new_route)
    return neighbors

def hill_climbing(n):
    distances = distance_matrix(n)
    route = list(range(n))
    random.shuffle(route)
    best_distance = total_distance(route, distances)
    print("Initial Route:", route)
    print("Initial Total Distance:", best_distance)
```

```
    while True:
        neighbors = generate_neighbors(route)
        new_route = min(neighbors, key=lambda r:
total_distance(r, distances))
        new_distance = total_distance(new_route, distances)

        if new_distance < best_distance:
            route, best_distance = new_route, new_distance
        else:
            break

    return route, best_distance

n = 5
route, best_distance = hill_climbing(n)
print("Best Route:", route)
print("Best Total Distance:", best_distance)
```

- Output:

```
PS D:\VIIT\SEM 4\AI\Lab> & C:/Python31
Initial Route: [1, 0, 4, 3, 2]
Initial Total Distance: 262
Best Route: [1, 0, 2, 4, 3]
Best total Distance: 180
PS D:\VIIT\SEM 4\AI\Lab>
```